

Henry Mashen Masanja
EARTH University, Costa Rica

May 2, 2026.

Prof. Christian Seiler,
School of Environmental Studies,
Queen's University
Kingston, Ontario, Canada.

Dear Prof. Seiler,

I am writing to express my strong interest in the MSc position investigating changes in land-atmosphere coupling in convection-permitting simulations. I am Henry Mashen Masanja, a fourth-year undergraduate at EARTH University in Costa Rica, graduating with a BSc in Agricultural Sciences and Engineering on 4 December 2026, with a cumulative GPA of 3.6/4.0. I will be available to begin graduate studies in January 2027 or immediately thereafter my graduation. Your research on soil moisture-precipitation feedback and land-atmosphere coupling under climate change represents a compelling intersection of environmental science, climate modeling, and data analysis, areas that deeply align with my research background and career aspirations.

My current BSc thesis, "Estimating Aboveground Biomass of Kernza Using UAV Multispectral Imagery and Machine Learning," has equipped me with strong technical competencies directly applicable to your MSc project, where I develop machine learning algorithms and statistical models in Python and R to extract patterns from complex multidimensional datasets, skills transferable to applying Granger causality and convergent cross mapping techniques for disentangling land-atmosphere interactions. I am experienced in processing, analyzing, and interpreting large-scale geospatial and environmental datasets, including handling missing data, temporal variability, and extreme events which are directly applicable to evaluating observed soil moisture and precipitation relationships. Additionally, my research quantifies the relationships between soil moisture, biomass, and land surface conditions across varying climatic and soil conditions, providing foundational understanding of land-surface-atmosphere interactions in agricultural contexts.

During my 2025 research internship at Aarhus University under Dr. Eusun Han, I gained comprehensive experience in analyzing large-scale observational and modeled datasets to understand environmental processes. I have working knowledge of Python and R for scientific computing, GIS analysis (QGIS, Pix4D) for spatial data handling, and statistical methods for time-series analysis and extreme event quantification. My background in Agricultural Sciences and Engineering provides essential knowledge of soil-water-plant-atmosphere interactions, moisture limitation, and evapotranspiration processes, central to understanding the land-atmosphere feedback your project investigates.

I am deeply motivated by research that addresses climate change impacts through rigorous analysis of climate models and observational data. Your MSc project's focus presents an exceptional opportunity to contribute to advancing climate risk assessment and adaptation planning for Northeast North America. I am very interested in working within your research group and collaborating with the broader team across the three institutions. I am a Ugandan and I will require a Canadian study permit.

Thank you for considering my application. I look forward to discussing how my background aligns with your MSc project's objectives.

Best regards,

Henry Mashen Masanja
Ugandan citizen | Will require Canadian study permit
EARTH University, Costa Rica